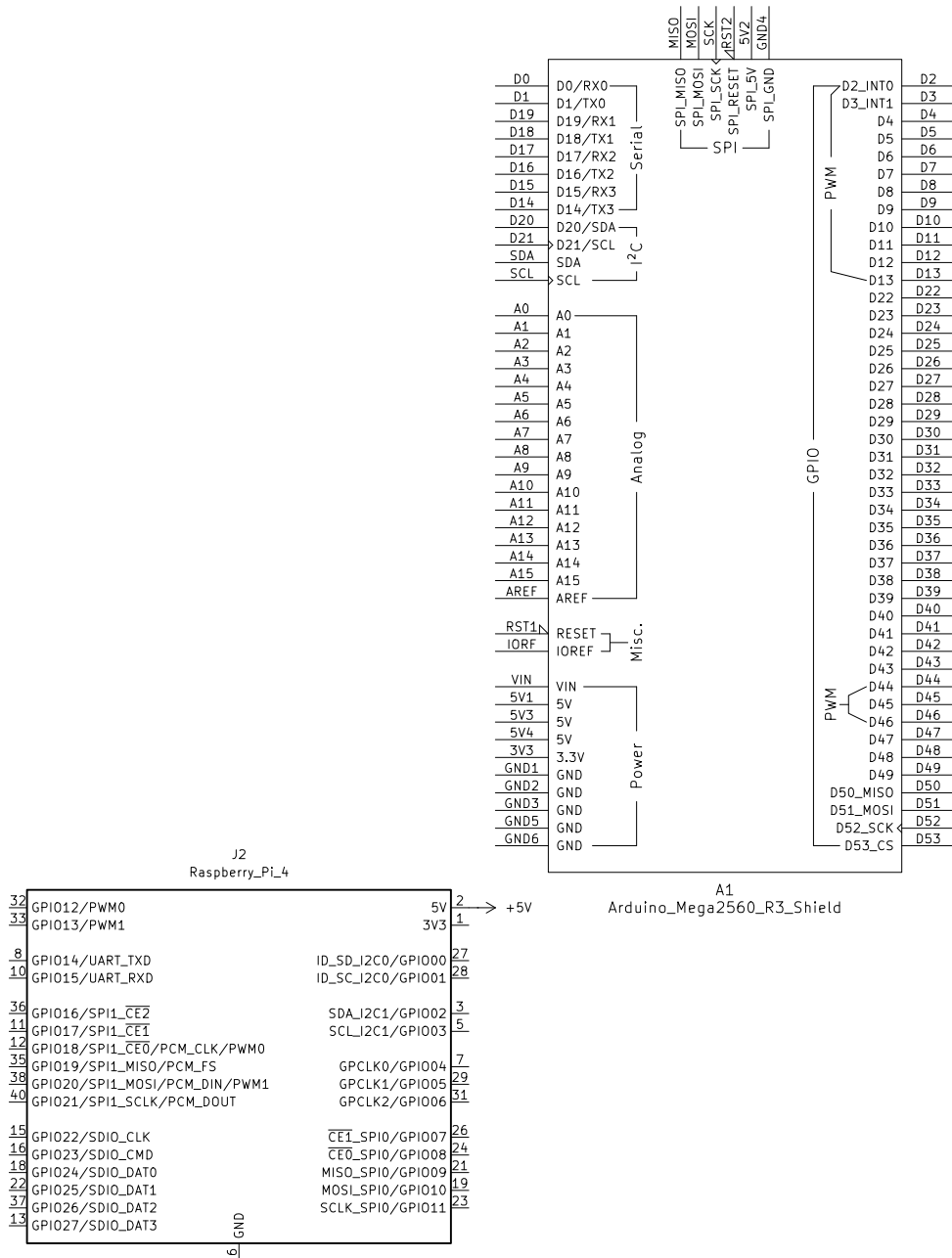
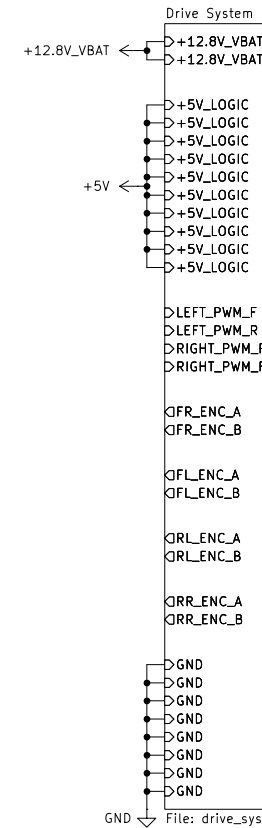


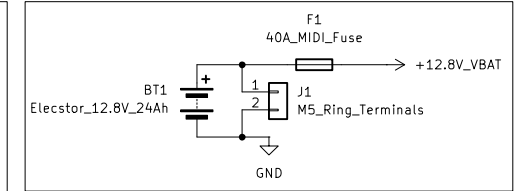


Pi Camera

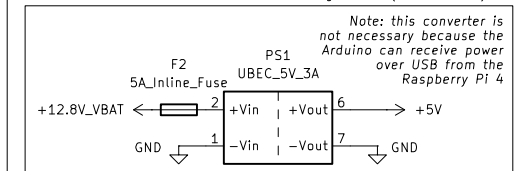
The Pi Camera circuitry is not shown; it connects to the Raspberry Pi 4 using a ribbon cable



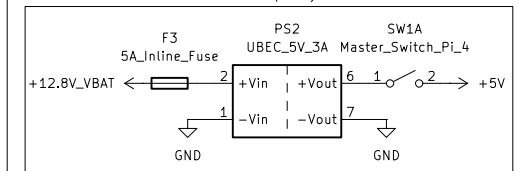
Battery Circuitry



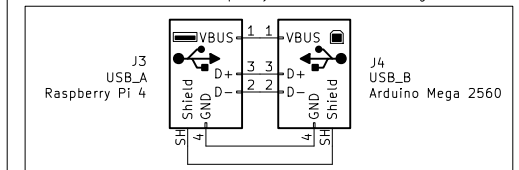
12.8V to 5V Buck Converter for Arduino Mega 2560 (NOT NEEDED)



12.8V to 5V Buck Converter for Raspberry Pi 4



Data and Power between Raspberry Pi 4 and Arduino Mega 2560



File: drive_system_wiring.kicad_sch

Pin 1: RPWM (Reverse Speed)
 Pin 2: LPWM (Forward Speed)
 Pin 3: R_EN (Reverse Enable)
 Pin 4: L_EN (Forward Enable)
 Pin 5: R_IS (Reverse Current Alarm – often left unconnected)
 Pin 6: L_IS (Forward Current Alarm – often left unconnected)
 Pin 7: VCC (5V Logic Power)
 Pin 8: GND (Logic Ground)

Power Pinouts:
 1: B+
 2: B-
 3: Motor +
 4: Motor -

Motor Pinouts:
 1: Motor +
 2: Motor -
 3: Hall_GND
 4: Hall_VCC_5V
 5: Hall_A_Out
 6: Hall_B_Out

